TOGAF

Enterprise Architecture Method and Governance Framework

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1 Overview

1.1 What is TOGAF

TOGAF is an enterprise architecture standard from The Open Group. It provides a structured method, content framework, and governance approach for designing, planning, implementing, and governing enterprise architecture.

TOGAF is not only a way to draw architecture diagrams. It is a complete approach for moving from business strategy to target architecture, transition roadmaps, implementation governance, and architecture capability development. It helps organisations align business, data, application, and technology change in a controlled and repeatable way.

1.2 Purpose

The purpose of TOGAF is to help enterprises:

- translate business strategy into architecture direction;
- design current-state and target-state enterprise architecture;
- govern change through a repeatable architecture development method;
- create transition roadmaps from baseline to target states;
- establish architecture principles, standards, and decision rights; and
- improve alignment across strategy, delivery, operations, risk, and technology platforms.

1.3 When to Use It

TOGAF is especially useful when an organisation needs a disciplined architecture method across business and technology change. Typical situations include:

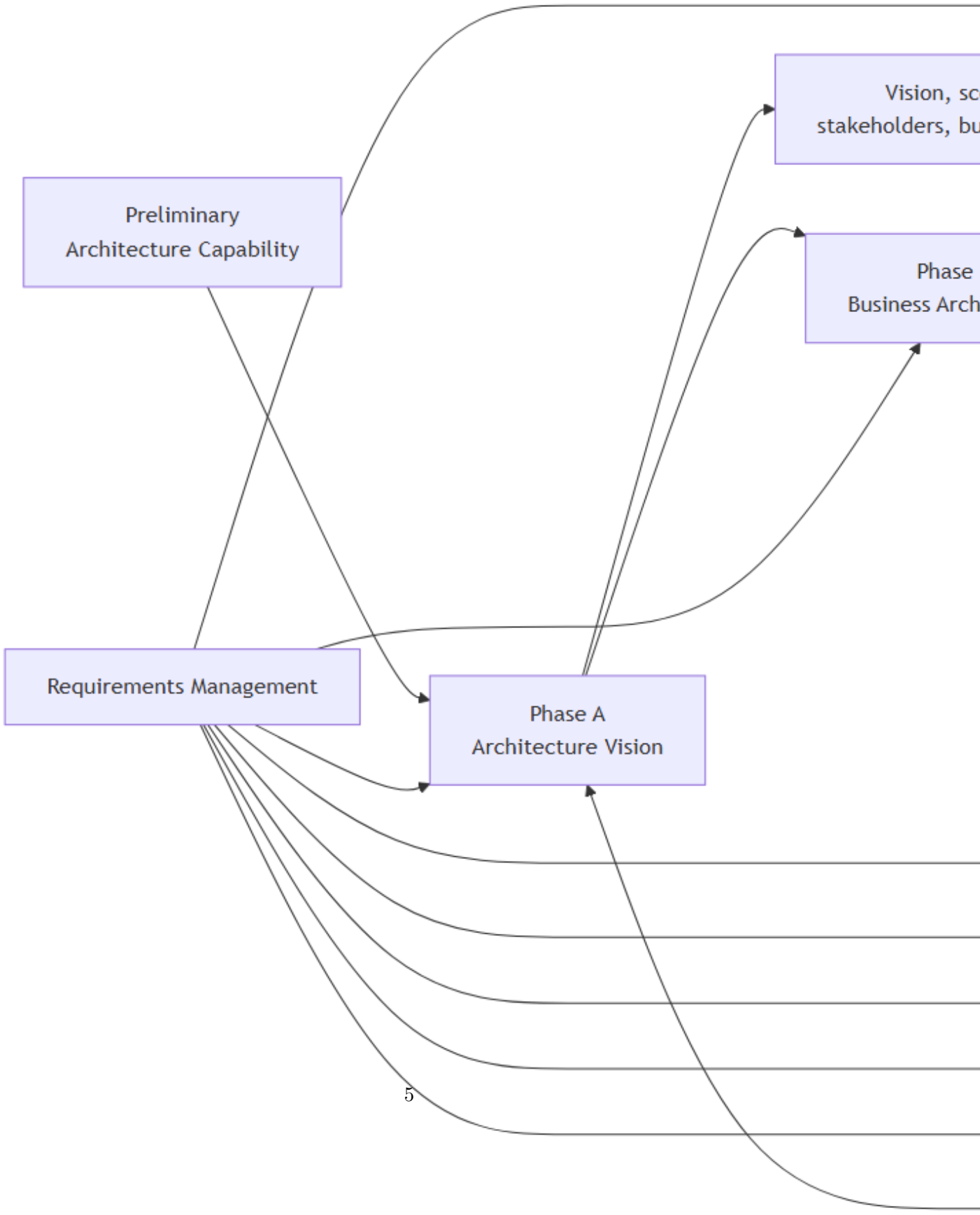
- enterprise transformation and operating model redesign;
- modernisation of applications, data platforms, or infrastructure;
- digital strategy execution and capability-based planning;

- mergers, acquisitions, or organisational restructuring;
- cloud migration and platform rationalisation;
- development of enterprise standards and architecture governance; and
- integration of multiple delivery programmes under one target-state design.

1.4 Target Audience

- CIO
- Chief Architect / Head of Architecture
- Enterprise Architects
- Solution Architects
- Business Architects
- Data and Application Architects
- PMO / Transformation Office
- Portfolio and Strategy teams
- Risk, security, and governance leaders
- Senior business stakeholders and design authorities

1.5 TOGAF Architecture Development Cycle



TOGAF is centred on the **Architecture Development Method (ADM)**. The ADM is iterative rather than strictly linear. Requirements are managed continuously, and architecture is refined over multiple cycles as the organisation learns, changes direction, or responds to constraints.

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2 History and Evolution

Version	Year	Key Changes
TOGAF 1	1995	Early framework focused on technical architecture
TOGAF 7	2001	Broader enterprise architecture scope and method development
TOGAF 8	2002–2003	Enterprise Edition with stronger business alignment
TOGAF 8.1.1	2006	Improved guidance and practical use
TOGAF 9	2009	Major expansion of ADM, content framework, governance, and repository
TOGAF 9.1	2011	Clarifications, better structure, and usability improvements
TOGAF 9.2	2018	Streamlining and content updates
TOGAF Standard, 10th Edition	2022	Modular structure with Fundamental Content and Series Guides for tailoring and practical guidance

TOGAF has evolved from a technical architecture method into a broad enterprise architecture standard. The major shift came with TOGAF 9, which established the modern ADM, content framework, enterprise continuum, repository, and governance approach. The 10th Edition made the standard more modular by separating stable core content from a growing set of detailed guidance documents.

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3 Core Principles

TOGAF does not prescribe one universal list of architecture principles for all organisations. Instead, it helps organisations define their own architecture principles and apply them consistently through governance. In practice, TOGAF is built around several enduring ideas:

1. **Business-driven architecture**
   Architecture should be anchored in business goals, stakeholder concerns, and strategic outcomes.
2. **Whole-enterprise perspective**
   Architecture should cover business, data, applications, and technology, not isolated technical components.
3. **Standardisation with controlled flexibility**
   Common principles, standards, and reusable building blocks reduce complexity while allowing local variation where justified.
4. **Iterative development**
   Architecture evolves through cycles, transition states, and managed change rather than one-time design exercises.
5. **Governed decision-making**
   Architecture must be supported by clear authority, review mechanisms, compliance processes, and escalation paths.
6. **Reuse and building blocks**
   Reusable architecture building blocks and solution building blocks accelerate delivery and improve consistency.
7. **Traceability from strategy to implementation**
   Decisions should connect stakeholder needs, architecture models, roadmaps, implementation work packages, and realised outcomes.

These principles matter because TOGAF works best when architecture is treated as an enterprise management capability rather than a documentation exercise.

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4 Governance Structure

4.1 Architecture Governance vs Delivery Governance

TOGAF distinguishes architecture governance from general project or operational governance:

- **Architecture governance** ensures that architecture principles, standards, target states, and design decisions are defined, reviewed, approved, and followed.
- **Delivery governance** ensures that projects and programmes are managed well, on time, and within scope.

This distinction is important because a project can be well-managed yet still produce a solution that increases technical debt, duplicates platforms, or conflicts with enterprise direction. TOGAF adds the architecture control layer.

4.2 Core Architecture Domains

Domain	Description
Business Architecture	Business strategy, capabilities, organisation, value streams, processes, services, and operating model design.
Data Architecture	Data entities, information flows, ownership, data lifecycle, quality, and data governance structures.
Application Architecture	Application services, integration, interfaces, interoperability, and application portfolio structure.
Technology Architecture	Infrastructure, platforms, cloud, networks, middleware, hosting, and technical standards.

4.3 Architecture Capability and Decision Bodies

A TOGAF-based architecture capability commonly includes:

- architecture board or design authority;
- chief architect and architecture practice;
- domain architects and solution architects;
- architecture repository and standards library;
- architecture principles and policies;
- review, waiver, and compliance processes; and
- linkage to portfolio, PMO, risk, security, data, and service management.

4.4 TOGAF Content Structure

The TOGAF Standard, 10th Edition is organised around two major content layers:

- **Fundamental Content**, which provides the essential framework and ADM structure; and
- **Series Guides**, which provide more specific guidance for applying TOGAF in different contexts.

This makes TOGAF more adaptable than earlier editions because organisations can keep a stable core while using more targeted guidance where needed.

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5 Processes

TOGAF is built around the Architecture Development Method (ADM).

5.1 Preliminary Phase

Phase	Purpose
Preliminary	Establish architecture capability, governance, principles, scope, and organisational readiness.

5.2 ADM Phases

Phase	Name	Typical Outputs
A	Architecture Vision	Statement of architecture work, stakeholders, scope, high-level target, and business case
B	Business Architecture	Capability maps, process models, organisational views, business services, value streams
C	Information Systems Architectures	Data architecture and application architecture models, integration views, information requirements

Phase	Name	Typical Outputs
D	Technology Architecture	Platform standards, infrastructure landscape, environments, technical services
E	Opportunities and Solutions	Work packages, solution concepts, transition options, dependencies
F	Migration Planning	Architecture roadmap, transition architectures, implementation sequencing
G	Implementation Governance	Architecture contracts, design reviews, compliance assessments, change controls
H	Architecture Change Management	Change requests, lifecycle monitoring, triggers for new ADM cycles

5.3 Central ADM Function

Function	Description
Requirements Management	Captures, assesses, prioritises, and manages requirements across all ADM phases.

5.4 Supporting TOGAF Concepts

TOGAF also includes several important supporting concepts that strengthen the ADM:

- **Architecture Principles** to guide consistent decisions;
- **Stakeholder Management** to align concerns and viewpoints;
- **Architecture Repository** to store standards, models, building blocks, and governance records;
- **Enterprise Continuum** to classify reusable assets from generic to organisation-specific;
- **Architecture Building Blocks (ABBs)** and **Solution Building Blocks (SBBs)**;
- **Capability-Based Planning** to connect transformation to business outcomes;
- **Gap Analysis** between baseline and target states; and
- **Architecture Compliance Reviews** during implementation.

6 Documentation and Artefacts

This section defines the key documents produced within the framework and the roles responsible for creating, approving, and maintaining them.

6.1 TOGAF Documentation RACI Matrix

Legend: R = Responsible, A = Accountable, C = Consulted, I = Informed

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Architecture Charter / Mandate	A	R	R	I	C	C	I
Architecture Principles	I	A	R	C	I	C	C
Architecture Governance Framework	I	A	R	C	C	C	I
Architecture Repository Structure	I	C	A/R	C	I	C	I
Architecture Vision	I	A	R	C	C	C	C

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Stakeholder Map and Concerns Matrix	I	C	A	R	C	C	C
Baseline Business Architecture	I	C	A	R	C	C	C
Target Business Architecture	I	C	A	R	C	C	C
Capability Map	I	C	A	R	C	C	C
Value Stream / Process Architecture	I	C	A	R	C	C	C
Baseline Data Architecture	I	C	A	R	I	C	C
Target Data Architecture	I	C	A	R	I	C	C

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Information Model / Data Catalogue	I	C	A	R	I	C	C
Baseline Application Architecture	I	C	A	R	I	C	C
Target Application Architecture	I	C	A	R	I	C	C
Integration Architecture / Interface Map	I	C	A	R	I	C	C
Baseline Technology Architecture	I	C	A	R	I	C	I
Target Technology Architecture	I	C	A	R	I	C	I

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Technology Standards Catalogue	I	C	A	R	I	C	I
Architecture Building Blocks Catalogue	I	C	A	R	I	C	I
Solution Building Blocks Catalogue	I	C	A	R	I	C	I
Gap Analysis	I	C	A	R	C	C	C
Opportunities and Solutions Assessment	I	A	R	C	C	C	C
Transition Architecture Definition	I	C	A	R	C	C	C

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Architecture Roadmap	I	A	R	C	R	C	C
Migration Plan	I	A	C	I	R	C	C
Architecture Contract	I	A	R	C	C	C	C
Architecture Compliance Assessment	I	C	A	R	I	C	I
Waiver / Dispensation Record	I	A	R	C	I	C	I
Architecture Decision Log	I	C	A	R	I	C	I
Architecture Change Request	I	C	A	R	C	C	C
Implementation Review Report	I	C	A	R	C	C	I

Document / Artefact	Board / Executive Sponsor	CIO	Chief / Enterprise Architect	Domain / Solution Architect	PMO / Transformation Office	Security / Risk / Data Governance	Product / Service / Business Owner
Lessons Learned Repository	I	C	A	C	R	C	I

7 Integration with Other Frameworks

Framework	Relationship
COBIT	Provides the enterprise governance layer, decision rights, controls, and performance oversight above architecture practice.
ITIL	Uses architecture outputs to shape service design, service transition, platform standards, and operational service models.
PRINCE2	Consumes architecture constraints, standards, and target-state guidance within project governance and stage delivery.
MSP	Uses target architectures and transition states to shape programme blueprints and transformation roadmaps.
PMBOK / Agile / DevOps	Delivery methods implement the solutions that architecture defines and governs.
ISO 27001 / NIST / security frameworks	Security requirements and controls should be embedded into business, data, application, and technology architecture decisions.
Data governance frameworks	Data ownership, metadata, quality, lifecycle, and information architecture align strongly with TOGAF data architecture work.

8 Implementation Approach

A practical TOGAF implementation usually works best when introduced as an architecture capability rather than only as a modelling method.

1. Assess current architecture maturity, governance gaps, and delivery pain points.
2. Define the architecture mandate, principles, scope, and decision rights.
3. Establish an architecture board, repository, standards catalogue, and review process.
4. Run ADM on one priority transformation area to prove value.
5. Build baseline and target architectures across the relevant domains.
6. Perform gap analysis and define work packages, transition architectures, and roadmap.
7. Integrate architecture governance into portfolio, project, product, and change forums.
8. Measure compliance, reuse, simplification, and business outcome contribution.

Organisations often fail when they start with excessive modelling. TOGAF tends to succeed when it is focused on concrete business change, supported by senior sponsorship, and linked directly to investment and delivery decisions.

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9 KPIs and Metrics

KPI	Measurement
Standards Compliance	% of solution designs compliant with approved architecture standards
Application Rationalisation	Reduction in duplicate or redundant applications
Technology Standardisation	% of platforms aligned to approved technology stack
Architecture Review Throughput	Number of architecture reviews completed per period
Waiver Volume	Number of exceptions granted and their risk profile
Reuse Rate	% of solutions using approved building blocks or reference architectures
Roadmap Delivery Alignment	% of change initiatives aligned to target architecture roadmap
Time to Approve Architecture Decisions	Average time from submission to architecture decision

KPI	Measurement
Technical Debt Reduction	Measured reduction in unsupported, duplicated, or non-standard components
Business Capability Coverage	% of priority capabilities mapped to target-state architecture

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10 Benefits

- Stronger alignment between strategy, operating model, and technology change
- Better decision-making through principles, standards, and governance
- Clearer target states and transition roadmaps for transformation
- Reduced duplication and technical debt through reuse and rationalisation
- Improved traceability from stakeholder needs to implemented solutions
- Greater consistency across business, data, application, and technology design
- Better collaboration between architects, delivery teams, PMO, risk, and operations

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11 Limitations

- Can become documentation-heavy if not tied to real transformation priorities
- Requires sustained executive backing and architecture authority
- Needs skilled architects and stakeholder engagement to be effective
- May be perceived as bureaucratic if reviews are slow or disconnected from delivery
- Does not replace delivery, service management, or governance frameworks; it complements them

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12 Example Use Case

A large enterprise is modernising its customer platforms, data estate, and core infrastructure while also moving toward a product operating model. Using TOGAF, it defines architecture principles, maps current capabilities, documents the baseline landscape, designs target business, data, application, and technology architectures, and creates phased transition architectures.

The roadmap then feeds portfolio prioritisation and project initiation, while architecture review boards ensure new solutions align with the target state and approved standards.

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13 References

- The Open Group, *TOGAF Standard*
- The Open Group, *TOGAF Standard, 10th Edition*
- The Open Group, TOGAF Series Guides
- Enterprise architecture governance and architecture practice guidance